

MD. RAKIB HOSSAIN

Full-Stack Developer | AI and Machine Learning

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Science and Engineering student who ships full-stack applications and machine-learning systems: a live e-commerce platform with a recommendation engine, a deployed ML risk service and a published hardware study. Uses AI-assisted development with Claude Code daily. Two papers accepted at ICCA 2026.

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript/JavaScript

Frontend: React, Next.js, Tailwind CSS, HTML/CSS

Backend: NestJS, Node.js, REST APIs, Prisma, PostgreSQL

AI & ML: PyTorch, scikit-learn, CatBoost, YOLO, OpenCV, pandas, NumPy

Tools: Git/GitHub, VS Code, Jupyter, Roboflow, Claude Code, Antigravity, Canva, Microsoft 365, Vercel, Render

PUBLICATIONS

- *DhakaNight: A Benchmark for Low-Light Object Detection in Dense Urban Night Traffic*. Accepted, ICCA 2026, Dhaka; ACM Digital Library.
- *PulseStone: A Tangible Single-Action Anxiety Companion*. Accepted, ICCA 2026, Dhaka.

PROJECTS

Drikon - Full-Stack E-Commerce Platform with Recommendation Engine | *Next.js 15, React 19, NestJS 11, PostgreSQL, Prisma*

- Implemented the Apriori association-rule algorithm from scratch - level-wise itemset mining with candidate pruning, scored by support, confidence and lift - to power "frequently bought together" and personalised recommendations from real order history.
- Built 17 backend modules - products, orders, reviews, coupons, flash sales, wishlists, banners, settings - with voice search, filtering, comparison and secure accounts on the storefront.
- Localised in Bengali and English, with branding, theme and homepage content editable from the admin panel without a redeploy.

[Live](#) | [Code](#)

RideGuard - Calibrated and Auditable ML Risk Framework | *Python, CatBoost, scikit-learn, SHAP*

- A live web app scoring motorcycle route risk in Dhaka, with a colour-coded risk map and real-time warnings for phone handling, speeding and accident hotspots.
- Predicts crash risk at 0.973 macro-F1, reporting its confidence and explaining each prediction rather than acting as a black box.
- A self-audit layer revealed the headline score came from a label the data gave away, which I published as the central finding.

[Research](#) | [App Code](#) | [Live](#)

DhakaNight - Night-Time Object Detection Dataset and Benchmark | *Python, PyTorch, YOLO, OpenCV, Roboflow*

- Detects vehicles and pedestrians on Dhaka streets after dark, where standard detectors fail.
- Labelled a 2,300-image night dataset, then benchmarked CLAHE, Gamma, Zero-DCE and RetinexFormer against a YOLOv8 baseline - none improved detection, and scaling the detector beat every pipeline.

[Code](#)

PulseStone - Handheld Anxiety-Relief Device | *ESP32-C3, Arduino C++, Python*

- A pocket device that starts a calming breathing exercise with one squeeze, guiding the user with light, vibration and screen cues.
- In a 15-participant, 150-trial study users started in 0.52 s versus 7.49 s on a phone app - 14.5x faster - all 15 preferred it.

[Code](#)

LEADERSHIP AND ACTIVITIES

Founder and Team Lead, "Khepa Chakka" Robotics Team

2023 - Present

- Founded and lead a competitive robotics team building four classes of robot: line-following, soccer, race and battle.
- Coordinate members across mechanical design, electronics and software, holding timelines against competition deadlines.

Active Member, Engineering Students' Association of Bangladesh (ESAB), AIUB2023 - Present

- Collaborate with faculty on research and development; participate in seminars and workshops on emerging technologies.

AWARDS AND HONORS

- Winner, Cozmo Clench - Techfest, IIT Bombay (Bangladesh Zonal), 2024. One of Asia's largest science and technology festivals.
- Champion, Robot Soccer - AIUB CS Fest, 2024.
- 1st Runner-up, Nano Battle Bot - National Robotics Championship, 2025.
- 2nd Runner-up, Robo Race - National Robotics Championship, 2025.
- 1st Runner-up, Robo Race - Technoxian Bangladesh National Round, 2024.
- 1st Runner-up, Robot Soccer - AIUB Robotic Crew (ARC).
- Competitor, BUET Autofest 2024 (Soccer Bot and Robo Race) and numerous national and inter-university robotics competitions.

EDUCATION

B.Sc. in Computer Science and Engineering, American International University-Bangladesh (AIUB)Expected 2026

CGPA: 3.28 / 4.00

Higher Secondary Certificate (HSC), Dhaka Imperial College2021

Secondary School Certificate (SSC), Dhaka Collegiate School2019

Languages: Bangla (Native), English (Fluent)